Abdur **Rouf**

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SKILLS

FRONT END DEVELOPMENT| React • Next.js • JavaScript (ES6+) • TypeScript • Tailwind CSS • DaisyUI • HTML • CSS
BACK END DEVELOPMENT| Node.js • Express.js • MongoDB • Firebase Auth • SQL • Flask
MACHINE LEARNING| Python • PyTorch • scikit-learn • NumPy • Pandas • TensorFlow Lite (INT8) • CNN • LSTM / GRU • Transformers • LangChain (RAG)
EMBEDDED SYSTEMS| Arduino • ESP32 • C / C++ • I2C • UART • Sensor integration • Circuit debugging
TOOLS| Git • GitHub • Figma • VS Code • Remote sensing & GIS
SOFT SKILLS| Team player • Fast learner • Problem solver • Calm under deadlines

EXPERIENCE

INDEPENDENT UNIVERSITY, BANGLADESH | Python, PyTorch, ESP32, Remote Sensing
Undergraduate Research | Dhaka, Bangladesh

Embedded thermal sensing with on-device machine learning

- Collected datasets with AMG8833 (8x8) and MLX90640 (32x24) thermal sensors for occupancy and posture classification.
- Trained a dual-head CNN for classification and heat-source localisation, quantised it to INT8 and deployed it on an ESP32.

National-scale PM2.5 exposure and brick-kiln vulnerability mapping

- Built a machine-learning framework estimating school-level PM2.5 exposure across Bangladesh.
- Fused satellite trace-gas retrievals, aerosol data and kiln proximity for 18k-34k geolocated institutions.

Smart farming hardware

- Multi-crop disease detection (cauliflower, strawberry, cabbage), spraying prevention and a farmer early-alert system.

Research produced 2 journal papers under review (Springer Nature, Elsevier) and 3 conference submissions (IEEE/ACM BDCAT 2026, SOCTA 2026).

PROGRAMMING HERO | React, Next.js, Node.js, MongoDB, Firebase
MERN Stack Web Development | 2025 - 2026 | Online

- Built responsive websites and React apps every week from Figma designs under strict deadlines.
- Shipped 7+ projects covering DOM scripting, React Router, charts, Firebase authentication and Next.js.

PROJECTS

Voltix Tech Store | Next.js, TypeScript, Firebase, Tailwind CSS

- E-commerce app with search and category, price and rating filters kept in the URL, plus dynamic product pages.
- Firebase email and Google sign-in with protected Add Product and Manage Products pages.

Flood Shield | React, Node.js, Express, MongoDB, Flask, XGBoost

- Flood monitoring and rescue platform with role-based pages for citizens, volunteers, NGOs and government.
- SOS alerts with a rescue queue, relief logistics, XGBoost flood risk, YOLOv8 detection and a RAG chatbot.

EDUCATION

INDEPENDENT UNIVERSITY, BANGLADESH (IUB)
B.Sc., Computer Science and Engineering
2022 - 2026 | Dhaka, Bangladesh
CGPA: 3.78/4.00

HONORS & AWARDS

TECHFEST CHAMPION | Cloud project: food donation platform | IUB | Spring 2024

TECHFEST CHAMPION & BEST HARDWARE PROJECT | Smart farming system | IUB Spring 2024, CARD-2025

TECHFEST CHAMPION | Algorithm project | IUB | Spring 2023

TECHFEST RUNNER-UP | Line-following robot | IUB | Summer 2023

VC'S LIST & DEAN'S LIST | IUB | Spring 2023, Summer 2022

PROBLEM SOLVING

LeetCode: 32 solved (26 Easy, 6 Medium)
HackerRank: Problem Solving (1 star), Java (3 stars)

CERTIFICATIONS

MERN Stack Developer: Programming Hero
Graphics and UI Designer: SEIP
CCNA Module 1: Cisco Networking Academy

COURSEWORK

Algorithms • Object-Oriented Programming (Java) • Digital Logic Design • Microprocessors • Numerical Methods • Data Mining

MORE PROJECTS

HERO IO APP STORE | React, React Router, Recharts. App store with search, rating charts and install tracking.

RESCUELINE BD | HTML, Tailwind, JavaScript. Emergency hotline directory with call history. Live on GitHub Pages.

LANGUAGES

English • Bangla • French (Beginner)

INTERESTS

Electronics • RC cars • CanSat • Tech fests • Learning French • Remote sensing research

PUBLICATIONS

A. Yasin, **A. Rouf** (corresponding author), et al. *Real-Time Joint Classification and Heat-Source Localization on Native 8x8 Thermal Frames with a Lightweight Dual-Head CNN for Microcontroller Deployment.* Scientific Reports (Springer Nature). **Under review**

M. Khan, S. Rashid, **A. Rouf** (equal contribution), et al. *Mapping School Vulnerability to Brick Kiln Exposure in Bangladesh Using Remote Sensing and Geospatial Analysis.* Remote Sensing Applications: Society and Environment (Elsevier). **Under review**

A. Rouf, et al. *A Novel Machine Learning-Based PM2.5 Exposure Estimation Framework for Educational Institutions in Bangladesh.* IEEE/ACM BDCAT 2026. **Submitted**

A. Rouf, et al. *An AI/ML-based Remote-Sensing Framework for Automatic Vulnerability Analysis of Educational Institutions Near Brick Kilns in Bangladesh.* IEEE/ACM BDCAT 2026. **Submitted**

S. M. Haa-mim, M. R. Uddin, **A. Rouf**, et al. *A Dual-Head Convolutional Network for Joint Human Presence, Posture and Bounding-Box Estimation from Ultra-Low-Resolution Thermal Images.* SOCTA 2026. **Submitted**

A. Rouf (first author). *MAE-Pretrained Multi-Scale Lightweight CNN for Four-Class Brain Tumor MRI Classification.* FICTA 2027. **In preparation**